

Microglia derive from progenitors, originating from the yolk sac, and which proliferate in the brain.

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Microglia, the resident CNS macrophages, represent about 10% of the adult brain cell population. Although described a long time ago, their origin and developmental lineage is still debated. While del Rio-Hortega suggested that microglia originate from meningeal macrophages penetrating the brain during embryonic development, many authors claim that brain parenchymal microglia derive from circulating blood monocytes originating from bone marrow. We have previously reported that the late embryonic and adult mouse brain parenchyma contains potential microglial progenitors [F. Alliot, E. Lecain, B. Grima, B. Pessac, Microglial progenitors with a high proliferative capacity in the embryonic and the adult mouse brain, Proc. Natl. Acad. Sci. U.S.A. 88 (1991) 1541-1545]. We now report that they can be detected in the brain rudiment from embryonic day 8, after their appearance in the yolk sac and that their number increases until late gestation. We also show that microglia appear during embryonic development and that their number increases steadily during the first two postnatal weeks, when about 95% of microglia are born. Finally, the main finding of this study is that microglia is the result of in situ proliferation, as shown by the high proportion of parenchymal microglial cells that express PCNA, a marker of cell multiplication, in embryonic and postnatal brain. Taken together, our data support the hypothesis that terminally differentiated brain parenchymal microglia are derived from cells originating from the yolk sac whose progeny actively proliferates in situ during development.

Engrafted parenchymal brain macrophages differ from microglia in transcriptome, chromatin landscape and response to challenge.

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Microglia are yolk sac-derived macrophages residing in the parenchyma of brain and spinal cord, where they interact with neurons and other glial. After different conditioning paradigms and bone marrow (BM) or hematopoietic stem cell (HSC) transplantation, graft-derived cells seed the brain and persistently contribute to the parenchymal brain macrophage compartment. Here we establish that graft-derived macrophages acquire, over time, microglia characteristics, including ramified morphology, longevity, radio-resistance and clonal expansion. However, even after prolonged CNS residence, transcriptomes and chromatin accessibility landscapes of engrafted, BM-derived macrophages remain distinct from yolk sac-derived host microglia. Furthermore, engrafted BM-derived cells display discrete responses to peripheral endotoxin challenge, as compared to host microglia. In human HSC transplant recipients, engrafted cells also remain distinct from host microglia, extending our finding to clinical settings. Collectively, our data emphasize the molecular and functional heterogeneity of parenchymal brain macrophages and highlight potential clinical implications for HSC gene therapies aimed to ameliorate lysosomal storage disorders, microgliopathies or general monogenic immuno-deficiencies.

Microglia: unique and common features with other tissue macrophages.

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Microglia are highly specialized tissue macrophages of the brain with dedicated functions in neuronal

development, homeostasis and recovery from pathology. Despite their unique localization in the central nervous system (CNS), microglia are ontogenetically and functionally related to their peripheral counterparts of the mononuclear phagocytic system in the body, namely tissue macrophages and circulating myeloid cells. Recent developments provided new insights into the myeloid system in the body with microglia emerging as intriguing unique archetypes. Similar to other tissue macrophages, microglia develop early during embryogenesis from immature yolk sac progenitors. But in contrast to most of their tissue relatives, microglia persist throughout the entire life of the organism without any significant input from circulating blood cells due to their longevity and their capacity of self-renewal. Notably, microglia share some features with short-lived blood monocytes to limit CNS tissue damage in pathologies, but only bone marrow-derived cells display the ability to become permanently integrated in the parenchyma. This emphasizes the therapeutic potential of bone marrow-derived microglia-like cells. Further understanding of both fate and function of microglia during CNS pathologies and considering their uniqueness among other tissue macrophages will be pivotal for potential manipulation of immune cell function in the CNS, thereby reducing disease burden. Here, we discuss new aspects of myeloid cell biology in general with special emphasis on the brain-resident macrophages and microglia.

Distinct origins, gene expression and function of microglia and monocyte-derived macrophages in CNS myelin injury and regeneration.

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Central nervous system (CNS) injury incurs a rapid innate immune response, including that from macrophages derived from endogenous microglia and circulating monocytes infiltrating the lesion site. One example of such injury is the demyelination observed in the autoimmune disease multiple sclerosis (MS), where macrophages are implicated in both myelin injury and regeneration. Although initially microglia and monocyte-derived macrophages were considered to have identical origins, gene expression, and function, recent advances have revealed important distinctions in all three categories and have caused a paradigm shift in view of their unique identity and roles. This has important consequences for understanding their individual contribution to neurological function and therapeutic targeting of these populations in diseases like MS. Here, we address the differences between CNS endogenous and exogenously-derived macrophages with a particular focus on myelin damage and regeneration.

The influence of environment and origin on brain resident macrophages and implications for therapy.

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Microglia are the tissue-resident macrophages of the brain and spinal cord. They are critical players in the development, normal function, and decline of the CNS. Unlike traditional monocyte-derived macrophages, microglia originate from primitive hematopoiesis in the embryonic yolk sac and self-renew throughout life. Microglia also have a unique genetic signature among tissue resident macrophages. Recent studies identify the contributions of both brain environment and developmental history to the transcriptomic identity of microglia. Here we review this emerging literature and discuss the potential implications of origin on microglial function, with particular focus on existing and future therapies using

bone-marrow- or stem-cell-derived cells for the treatment of neurological diseases.

[Microglia arise from extra-embryonic yolk sac primitive progenitors].

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Microglia are the resident macrophage population of the central nervous system (CNS). Adequate microglia function is crucial for the homeostasis of the CNS in health and disease, as they represent the first line of defence against pathogens, contributing to immune responses, but are also involved in tissue repair and remodeling. It is therefore crucial to better understand microglia origin and homeostasis. Much controversy remains regarding the nature of microglial progenitors, as the exact contribution and persistence of embryonic and post-natal hematopoietic progenitors to the adult microglial pool in the steady state remained unclear. In this study, we show that post-natal hematopoietic progenitors do not significantly contribute to microglia homeostasis in the adult brain in mice. In vivo lineage tracing studies established that adult microglia derives from primitive hematopoietic progenitors that arise before embryonic day 8. These results identify microglia as an ontogenically distinct population in the mononuclear phagocyte system and have implications for the use of embryonically-derived microglial progenitors for the treatment of various brain disorders.

Most Tissue-Resident Macrophages Except Microglia Are Derived from Fetal Hematopoietic Stem Cells.

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Macrophages are one of the most diverse cell populations in terms of their anatomical location and functional specialization during both homeostasis and disease. Although it has been shown in different fate mapping models that some macrophages present in adult tissues are already established during fetal development, their exact origins are still under debate. In the current study, we developed a fate mapping strain, based on the Kit locus, which allowed us to readdress "the origins" question. Different types of macrophages from various adult tissues were traced to their fetal or adult sources by inducing labeling in precursors at several time points either during fetal development or in adult mice. We show that all adult macrophages, resident or infiltrating, are progenies of classical hematopoietic stem cells (HSC) with the exception of microglia and, partially epidermal Langerhans cells, which are yolk sac (YS)-derived.

A lineage of myeloid cells independent of Myb and hematopoietic stem cells.

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Macrophages and dendritic cells (DCs) are key components of cellular immunity and are thought to originate and renew from hematopoietic stem cells (HSCs). However, some macrophages develop in the embryo before the appearance of definitive HSCs. We thus reinvestigated macrophage development. We found that the transcription factor Myb was required for development of HSCs and all CD11b(high) monocytes and macrophages, but was dispensable for yolk sac (YS) macrophages and for the

development of YS-derived F4/80(bright) macrophages in several tissues, such as liver Kupffer cells, epidermal Langerhans cells, and microglia--cell populations that all can persist in adult mice independently of HSCs. These results define a lineage of tissue macrophages that derive from the YS and are genetically distinct from HSC progeny.

Ontogeny of Tissue-Resident Macrophages.

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The origin of tissue-resident macrophages, crucial for homeostasis and immunity, has remained controversial until recently. Originally described as part of the mononuclear phagocyte system, macrophages were long thought to derive solely from adult blood circulating monocytes. However, accumulating evidence now shows that certain macrophage populations are in fact independent from monocyte and even from adult bone marrow hematopoiesis. These tissue-resident macrophages derive from sequential seeding of tissues by two precursors during embryonic development. Primitive macrophages generated in the yolk sac (YS) from early erythro-myeloid progenitors (EMPs), independently of the transcription factor c-Myb and bypassing monocytic intermediates, first give rise to microglia. Later, fetal monocytes, generated from c-Myb(+) EMPs that initially seed the fetal liver (FL), then give rise to the majority of other adult macrophages. Thus, hematopoietic stem cell-independent embryonic precursors transiently present in the YS and the FL give rise to long-lasting self-renewing macrophage populations.

Human Induced Pluripotent Stem Cell-Derived Macrophages Share Ontogeny with MYB-Independent Tissue-Resident Macrophages.

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Tissue-resident macrophages, such as microglia, Kupffer cells, and Langerhans cells, derive from Myb-independent yolk sac (YS) progenitors generated before the emergence of hematopoietic stem cells (HSCs). Myb-independent YS-derived resident macrophages self-renew locally, independently of circulating monocytes and HSCs. In contrast, adult blood monocytes, as well as infiltrating, gut, and dermal macrophages, derive from Myb-dependent HSCs. These findings are derived from the mouse, using gene knockouts and lineage tracing, but their applicability to human development has not been formally demonstrated. Here, we use human induced pluripotent stem cells (iPSCs) as a tool to model human hematopoietic development. By using a CRISPR-Cas9 knockout strategy, we show that human iPSC-derived monocytes/macrophages develop in an MYB-independent, RUNX1-, and SPI1 (PU.1)-dependent fashion. This result makes human iPSC-derived macrophages developmentally related to and a good model for MYB-independent tissue-resident macrophages, such as alveolar and kidney macrophages, microglia, Kupffer cells, and Langerhans cells.